Week 1 Solutions

(1) (a) You are given $p(x,y)$ and you are supposed to find Bayes optimal. With squared loss, this is:

$$f^*(x) = \underset{Y/x \sim p(y/x)}{E}\left[Y/x\right]$$

So the steps are: (a) Find posterior $p(y/x)$ from $p(x,y)$
(b) Find mean of posterior

$$\text{Now, } p(y/x) = \frac{p(x,y)}{p(x)} \propto e^{-\frac{1}{2}\begin{bmatrix}x^T-\mu_1^T & y^T-\mu_2^T\end{bmatrix}\begin{bmatrix}\Lambda_{11} & \Lambda_{12}\\ \Lambda_{12}^T & \Lambda_{22}\end{bmatrix}\begin{bmatrix}x-\mu_1\\ y-\mu_2\end{bmatrix}}$$

$p(x)$ — Normalization const. for $p(y/x)$

$$\propto e^{-\frac{1}{2}(y-\mu_2)^T\Lambda_{22}(y-\mu_2) - (y-\mu_2)^T\Lambda_{12}^T(x-\mu_1)}$$

→ ignoring constants (terms not having "y")

$$\propto e^{-\frac{1}{2}\left(y-\mu_2+\Lambda_{22}^{-1}\Lambda_{12}^T(x-\mu_1)\right)^T\Lambda_{22}\left(y-\mu_2+\Lambda_{22}^{-1}\Lambda_{12}^T(x-\mu_1)\right)}$$

→ completing squares.

$$\Longrightarrow\quad Y/x \sim N\left(\mu_2-\Lambda_{22}^{-1}\Lambda_{12}^T(x-\mu_1),\ \Lambda_{22}^{-1}\right)$$

$$\therefore E\{Y/x\} = \mu_2-\Lambda_{22}^{-1}\Lambda_{12}^T(x-\mu_1)$$

Also see → (3.28) in Murphy book.

Now, we wish to re-write Λ in terms of Σ.
This can be done using Schur Complement (7.121) in Murphy book:

$$\Lambda_{22}^{-1} = \left(\Sigma_{22} - \Sigma_{12}^T \Sigma_{11}^{-1} \Sigma_{12} \right)$$

$$\Lambda_{12}^T = - \Lambda_{22} \Sigma_{21}^T \Sigma_{11}^{-1}$$

$$\Rightarrow E\{Y/x\} = \mu_2 + \underline{\Sigma_{21}^T \Sigma_{11}^{-1}} (x - \mu_1)$$

Note that the linear part of this affine function is same as that in Linear Regression!

(1b) In this case, mode of posterior is the Bayes optimal.

$$f^*(x) = \begin{cases} 1 & \text{if } p^*(1/x) \geq p^*(0/x) \\ 0 & \text{else.} \end{cases}$$

$$p^*(1/x) \geq p^*(0/x) \iff \frac{1}{1+e^{x^2-5x+6}} \geq 1 - \frac{1}{1+e^{x^2-5x+6}}$$

$$\iff x^2 - 5x + 6 \leq 0$$

$$\iff (x-3)(x-2) \leq 0$$

$$\iff \cancel{x \in (-\infty, 2] \cup [3, \infty)}$$

$$x \in [2, 3]$$

$$\therefore f^*(x) = \begin{cases} 1 & \text{if } x \in [2,3] \\ 0 & \text{else.} \end{cases}$$

(1c) $f^*(x) = \underset{y \in \{0,1\}}{\operatorname{argmin}} \; E[\ell(y, Y) | x]$

$= \underset{y \in \{0,1\}}{\operatorname{argmin}} \; \ell(y, 0)\left(1 - \frac{1}{1+e^{x^2-5x+6}}\right) + \ell(y,1)\left(\frac{1}{1+e^{x^2-5x+6}}\right)$

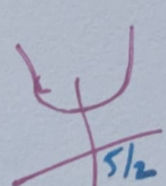
if $y=0$, then, $= \dfrac{0.5}{1+e^{x^2-5x+6}}$ (A)

if $y=1$, then $= 2\left(1 - \dfrac{1}{1+e^{x^2-5x+6}}\right)$ (B)

$\therefore f^*(x) = \begin{cases} 0 & \text{if } (A) \leq (B) \\ 1 & \text{else.} \end{cases}$

(A) $\leq$ (B) $\Leftrightarrow \dfrac{0.5}{1+e^{x^2-5x+6}} \leq 2\left(1 - \dfrac{1}{1+e^{x^2-5x+6}}\right)$

$\Leftrightarrow e^{x^2-5x+6} \geq \dfrac{1}{4}$



$\Leftrightarrow x^2 - 5x + (6 + \log 4) \geq 0$

roots $\rightarrow$ $\cancel{\dfrac{5 \pm \sqrt{25 - 24 - 4\log 4}}{2}}$ $\rightarrow$ imaginary roots

$\Leftrightarrow x \in \mathbb{R}.$

$\therefore f^*(x) = 0 \quad \forall x \in \mathbb{R}.$

1d $p^*(y/x) \propto e^{-\Lambda(x)y}, \quad y > 0$

$\longrightarrow$ exponential distribution with ~~mean as $\Lambda(x)$~~. parameter as $\Lambda(x)$.

$$\Rightarrow E[Y/x] = \frac{1}{\Lambda(x)}$$

If you don't know, ~~then~~ then:

First find the normalization const: $\int_0^\infty e^{-\Lambda(x)y}\,dy$

$$\int_0^\infty e^{-\Lambda(x)y}\,dy = \frac{e^{-\Lambda(x)y}}{-\Lambda(x)}\Bigg|_0^\infty = \frac{1}{\Lambda(x)}.$$

$$\therefore\ p^*(y/x) = \Lambda(x)e^{-\Lambda(x)y}$$

$$E[Y/x] = \int_0^\infty y\,p^*(y/x)\,dy = \Lambda(x)\int_0^\infty y e^{-\Lambda(x)y}\,dy$$

$$= \Lambda(x)\left[\frac{y e^{-\Lambda(x)y}}{-\Lambda(x)}\Bigg|_0^\infty + \int_0^\infty \frac{e^{-\Lambda(x)y}}{\Lambda(x)}\,dy\right]$$

By parts

$$= \Lambda(x)\left[\frac{1}{\Lambda(x)^2}\right] = \frac{1}{\Lambda(x)}.$$

1e $f^*(x) = \underset{y>0}{\operatorname{argmin}}\ E\big[|y - Y|\,/x\big]$

$$= \underset{y>0}{\operatorname{argmin}}\ \underbrace{\int_0^y (y-y')\,p(y'/x)\,dy' + \int_y^\infty (y'-y)\,p(y'/x)\,dy'}_{g(y)}$$

Differentiate $g(y)$ by LEIBNIZ rule:

$$g'(y) = 0 - 0 + \int_0^y p(y'/x)\,dy' + 0 - 0 - \int_y^\infty p(y'/x)\,dy'$$

$$g'(y) = 0 \iff \int_0^y p(y'/x)\,dy' = \int_y^\infty p(y'/x)\,dy'$$

$\iff$ y is median of $p(y/x)$.

$$g''(y) = 2p(y/x) \geq 0 \quad \forall y > 0$$

$\therefore$ Median is/are the only minimizer(s).

From (1d), posterior is exponential.

$$\therefore f^*(x) = \frac{\ln 2}{\lambda(x)}.$$

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if you don't ~~know~~,

Solve: $$\int_0^y e^{-\lambda(x)y'}\,dy' = \int_y^\infty e^{-\lambda(x)y'}\,dy'$$

$$\iff \left.\frac{e^{-\lambda(x)y'}}{-\lambda(x)}\right|_0^y = \left.\frac{e^{-\lambda(x)y'}}{-\lambda(x)}\right|_y^\infty$$

$$\iff \frac{e^{-\lambda(x)y}}{-\lambda(x)} + \frac{1}{\lambda(x)} = \frac{e^{-\lambda(x)y}}{\lambda(x)} \iff e^{-\lambda(x)y} = \frac{1}{2}$$

$$\iff y = \frac{\ln 2}{\lambda(x)}.$$

Week 2 Solutions

(1) $$\min_{w \in \mathbb{R}^n} \sum_{i=1}^{m} |w^T x_i - y_i|$$

$$= \min_{w \in \mathbb{R}^n,\, t \in \mathbb{R}^m} \sum_{i=1}^{m} t_i$$

s.t. $|w^T x_i - y_i| \le t_i \quad \forall\, i = 1 \text{ to } m$

$$\min_{x, z} z \quad \text{s.t. } f(x) \le z \;=\; \min_x \boxed{\begin{matrix} \min_z z \\ \text{s.t. } z \ge f(x) \end{matrix}} \;=\; \min_x f(x)$$

$$\boxed{\begin{matrix} \min_z z \\ \text{s.t. } z \ge f(x) \end{matrix}} = f(x)$$

$$= \min_{w \in \mathbb{R}^n,\, t \in \mathbb{R}^m} \sum_i t_i$$

s.t.
$w^T x_i - y_i \le t_i \quad \forall\, i = 1 \text{ to } m$
$y_i - w^T x_i \le t_i \quad \forall\, i = 1 \text{ to } m$

Linear Program.

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(2) $\sum_{i=1}^{m} x_i x_i^T$ is invertible $\iff$ $x_1 \ldots x_m$ span $\mathbb{R}^n$

($x_i \in \mathbb{R}^n$)

$\sum_{i=1}^{m} x_i x_i^T$ is invertible

$\iff \exists\, B \ni B \sum_{i=1}^{m} x_i x_i^T = \sum_{i=1}^{m} x_i x_i^T B = I.$ — (A)

$x_1, \ldots x_m$ span $\mathbb{R}^n \iff$ given any $z \in \mathbb{R}^n$, $\exists\, y \ni$

$$\sum_{i=1}^{m} x_i y_i = z.$$

$\Longrightarrow$ Proof by construction

Let $z \in \mathbb{R}^n$ be any given vector.

LHS $\Leftrightarrow$ (A) $\Rightarrow B \sum_{i=1}^{m} x_i x_i^T = I$

$$\Rightarrow B \sum_{i=1}^{m} x_i x_i^T B z = B z I = B z$$

$$\Rightarrow \left(\sum_{i=1}^{m} x_i x_i^T\right) B \sum_{i=1}^{m} x_i x_i^T B z = \left(\sum_{i=1}^{m} x_i x_i^T\right) B z$$

$$\Rightarrow \sum_{i=1}^{m} x_i x_i^T B z = z \quad \left(\because \sum_{i=1}^{m} x_i x_i^T B = I \text{ from (A)}\right)$$

$$\Rightarrow \sum_{i=1}^{m} x_i y_i = z, \text{ where } y_i = x_i^T B z.$$

$\Leftrightarrow$ RHS

$\Longleftarrow$ Proof by construction

RHS $\Rightarrow \exists\, y_i \ni X y_i = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow i^{th}$ position, $i = 1$ to n

$\Rightarrow XY = I$, where $Y = [y_1 \cdots y_n]$

~~RHS~~ $\Rightarrow y_i^T X^T = [0..010..0] \rightarrow i^{th}$ position, $i = 1$ to n.

$\Rightarrow Y^T X^T = I.$

$XY = I \Rightarrow X$

$\Leftarrow$ (Duality result)

$x_1, x_2, \ldots, x_m$ span $\mathbb{R}^n$

W.l.o.g, $\underbrace{x_1, \ldots, x_n}_{\text{basis}}, \underbrace{x_{n+1} \ldots x_m}_{\text{rest}}$ ($\because$ Every spanning set has a basis)

$$X = [\underbrace{x_1 \cdots x_n}_{\downarrow} \; x_{n+1} \cdot x_m]$$

$$= [\bar{X} \quad \bar{X}\Lambda]$$

$$\Rightarrow XX^T = [\bar{X} \quad \bar{X}\Lambda]\begin{bmatrix} \bar{X}^T \\ (\bar{X}\Lambda)^T \end{bmatrix} = \bar{X}\bar{X}^T + \bar{X}\Lambda\Lambda^T\bar{X}^T$$

$$\Rightarrow v^T X X^T v = v^T \bar{X}\bar{X}^T v + \underbrace{v^T \bar{X}\Lambda\Lambda^T\bar{X}^T v}_{\geq 0}$$

$$\geq v^T \bar{X}\bar{X}^T v$$

$$= (\bar{X}^T v)^2$$

> 0 whenever $v \neq 0$.

Because $\rho(\bar{X}) = \mathbb{R}^n \therefore N(\bar{X}^T) = \{0\}$ ($\because$ Rank-nullity theorem)

$\Rightarrow$ $XX^T v \neq 0$ if $v \neq 0$

$\Rightarrow$ $N(XX^T) = \{0\}$ ($\because$ Rank Nullity theorem)

$\Rightarrow$ $\rho(XX^T) = \mathbb{R}^n \Rightarrow \underset{\exists Y \ni}{XX^T Y = I}$

(2) $$\min_{W \in \mathbb{R}^{n \times d}} E\left[\| Y - W^T \varphi(x) \|_{\ell}^2 \right]$$

$$= \min_{W \in \mathbb{R}^{n \times d}} E\left[\sum_{i=1}^{d} \underbrace{\left(w_i^T \varphi(x) - y_i \right)^2}_{\to\, W = [w_1 \cdots w_d]} \right]$$

$$= \min_{w_1 \in \mathbb{R}^n, \ldots, w_n \in \mathbb{R}^n} \sum_{i=1}^{d} E\left[\left(w_i^T \varphi(x) - Y_i \right)^2 \right]$$

$$= \sum_{i=1}^{d} \underbrace{\min_{w_i \in \mathbb{R}^n} E\left[\left(w_i^T \varphi(x) - Y_i \right)^2 \right]}$$

$\to$ classical linear regression.

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(3) $$\ell_\Sigma\left(Y, W^T \varphi(x)\right) = \left(Y - W^T(\varphi(x))\right)^T \Sigma \left(Y - W^T \varphi(x)\right)$$

$$= \| \bar{Y} - \bar{W}^T \varphi(x) \|^2, \text{ where } \bar{Y} = \Sigma^{1/2} Y$$

$$\bar{W} = W \Sigma^{1/2}$$

Week 3 Solutions

(1) (a) Solving directly is cumbersome.

Let us look at Bayes optimal:

$$f^*(x) = 1 \text{ iff } p(1/x) \geq p(-1/x) \quad , x \in [-3,3]$$

$$\Leftrightarrow \frac{(x-3)^2}{36} \geq \frac{1}{2} \quad , x \in [-3,3]$$

$$\Leftrightarrow x \in [-3, 3-3\sqrt{2}]$$

$$\therefore f^*(x) = \text{sign}(3-3\sqrt{2}-x) \quad , x \in [-3,3].$$

$$= \text{sign}(w^{*T}\varphi(x)) \quad , x \in [-3,3]$$

$$\text{where } w^* = \begin{bmatrix} -1 \\ 3-3\sqrt{2} \end{bmatrix}$$

$\therefore$ w^* must be defining Bayes optimal linear with 0-1 loss

$\longrightarrow$ Solving for case of hinge loss is cumbersome.

Main point is w^* need not be minimizer.

$\hookrightarrow$ To show this let us assume $w = \begin{bmatrix} -1 \\ w_2 \end{bmatrix}$ we seek such solutions. Then,

$$\min_{w_2 \in \mathbb{R}} E\left[\max\left(0, 1 - Y(-X + w_2)\right)\right]$$

$$= \min_{w_2 \in \mathbb{R}} \int_{-3}^{3} \max(0, 1 + x - w_2)\, p(1/x)\, p(x)\, dx$$

$$+ \int_{-3}^{3} \max(0, 1 - x + w_2)\, p(-1/x)\, p(x)\, dx$$

$$= \frac{1}{216} \min_{w_2 \in \mathbb{R}} \underbrace{\int_{w_2 - 1}^{3} (1 + x - w_2)(x-3)^2\, dx}_{g_1(w_2)}$$

$$+ \underbrace{\int_{-3}^{1 + w_2} (1 - x + w_2)\left(36 - (x-3)^2\right) dx}_{g_2(w_2)}$$

By ~~Leibnitz~~ Leibniz rule,

$$\frac{d g_1(w_2)}{d w_2} = -\left(1 + (w_2 - 1) - w_2\right)(w_2 - 1 - 3)^2 \cdot 1 + \int_{w_2 - 1}^{3} -(x-3)^2\, dx$$

$$= \frac{(w_2 - 4)^3}{3}$$

IIIly,

$$\frac{d g_2(w_2)}{d w_2} = \int_{-3}^{1 + w_2} \left(36 - (x-3)^2\right) dx = (w_2 + 4)36 - \frac{(w_2 - 2)^3}{3} - 72$$

$$\therefore \frac{d}{dw_2}\left(g_1(w_2) + g_2(w_2)\right) = \frac{(w_2-2-2)^3}{3} - \frac{(w_2-2)^3}{3} + 36w_2 + 72$$

$$= -\frac{8}{3} - (w_2-2)^2 2 + (w_2-2)4 + 36w_2 + 72$$

$$= -w_2^2 + 32w_2 + \frac{160}{3}$$

$$\hat{w}_2 = \frac{-32 \pm \sqrt{32^2 + 640/3}}{-2}$$

$$= 16 \pm \sqrt{3712/3} \neq w_2^*.$$

Along similar lines, using Leibniz rule, compute Bayes linear classifier with hinge loss.

(b) For 0-1 loss, since ~~[illegible]~~ Bayes optimal itself is a linear classifier, Model error $= 0$.

For hinge-loss, Bayes optimal is defined by:

$$f^*(x) = \underset{y \in \{-1,1\}}{\arg\min} \; E[\max(0, 1-yY)/x]$$

$$E[\max(0,1-Y)/x] = \begin{cases} 2\, p(-1/x) \end{cases} \quad \therefore \text{Bayes optimal is the same}$$

$$E[\max(0,1+Y)/x] = 2\, p(1/x)$$

Bayes linear classifier with risk

Once (a) is solved, risk is computed, $\mathcal{R}(x)$

So model error is $\mathcal{R}(x) - 2\min\left(p(-1/x), p(1/x)\right)$

(c) For 0-1 loss:

$f^*(x) = 1$ iff $p(1/x) \geq p(-1/x)$, $x \in [-B, B]$

$\iff \dfrac{(x-3)^2}{(B+3)^2} \geq \dfrac{1}{2}$, $x \in [-B, B]$

$\cancel{\iff x \in [B, 3}$

roots are $3 \pm \dfrac{(B+3)}{\sqrt{2}}$

$\iff x \in \left[-B, 3 - \dfrac{B+3}{\sqrt{2}}\right] \cup \cancel{\& \, x \in} \left[3 + \dfrac{B+3}{\sqrt{2}}, B\right]$

sign(quadratic)

$x \in \left[-B, 3 - \dfrac{B+3}{\sqrt{2}}\right]$

sign(affine)

if $B \leq 9 - 6\sqrt{2}$ or $B \geq 9 + 6\sqrt{2}$

$\iff$ else

$\therefore$ No model error iff $B \in [9 - 6\sqrt{2}, 9 + 6\sqrt{2}]$

In case of hinge-loss, model error need not be zero as (a) shows. ~~Entora~~

(d) For 0-1 loss:

Let's solve: $\min_{y \in \{-1,1\}} E\left[\mathbb{1}\{Y \neq y\}\right]$ Constant predictor

$$= \min\left(P(1), P(-1)\right)$$

$$= \min\left(\int P(1/x)P(x)\,dx, \int P(-1/x)P(x)\,dx\right)$$

$$= \min\left(\int_{-B}^{B} \frac{(x-3)^2}{(B+3)^2}\frac{1}{2B}\,dx, \int_{-B}^{B}\left(1-\frac{(x-3)^2}{(B+3)^2}\right)\frac{1}{2B}\,dx\right)$$

$$= \min\left(\frac{B^2+27}{3(B+3)^2}, 1-\frac{B^2+27}{3(B+3)^2}\right)$$

$$= \begin{cases} \dfrac{B^2+27}{3(B+3)^2} & \text{if } B \in [9-6\sqrt{3}, 9+6\sqrt{3}] \\ \dfrac{2B(B+9)}{3(B+3)^2} & \text{else.} \end{cases} \quad \text{(I)}$$

Let compute Bayes optimal and $B \notin (9-6\sqrt{2}, 9+6\sqrt{2})$

$$= \int_{-B}^{3-\frac{B+3}{\sqrt{2}}}\left(1-\frac{(x-3)^2}{(B+3)^2}\right)\frac{1}{2B}\,dx + \int_{3+\frac{B+3}{\sqrt{2}}}^{B}\frac{(x-3)^2}{(B+3)^2}\frac{1}{2B}\,dx$$

$$= \frac{1}{B}\left(\frac{12\sqrt{2}+21}{12\sqrt{2}}\right) + \frac{4\sqrt{2}-5}{12\sqrt{2}} \quad \text{(II)}$$

When is I = II ?

If $B \in (0, 9-6\sqrt{3}]$, or $B \in [9+6\sqrt{3}, \infty)$,

$$\frac{2B(B+9)}{3(B+3)^2} = \frac{1}{B}\left(\frac{12\sqrt{2}+21}{12\sqrt{2}}\right) + \frac{4\sqrt{2}-5}{12\sqrt{2}}$$

Solve!

If $B \in [9-6\sqrt{3}, 9-6\sqrt{2}]$ or $B \in [9+6\sqrt{2}, 9+6\sqrt{3}]$

$$\frac{B^2+27}{3(B+3)^2} = \frac{1}{B}\left(\frac{12\sqrt{2}+21}{12\sqrt{2}}\right) + \frac{4\sqrt{2}-5}{12\sqrt{2}}$$

Solve!

~~If~~ $\cancel{B \in [9+6\sqrt{2}, 9+6\sqrt{3}]}$

$$\cancel{\frac{B^2+27}{3(B+2)^2} =}$$

~~If~~ $\cancel{1}$

(2) $p(x/y) \propto \dfrac{e^{-\frac{1}{2}(x-\mu_y)^T \Sigma_y^{-1}(x-\mu_y)}}{|\Sigma_y|^{1/2}}$, $p(y) = \begin{cases} 0.4 & y=1 \\ 0.6 & y=-1 \end{cases}$

$p(y/x) \propto p(x/y)p(y)$

$\Rightarrow p(1/x) \propto 0.4 \dfrac{e^{-\frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1)}}{|\Sigma_1|^{1/2}}$

$f^*(x) = 1 \iff p(1/x) \geq p(-1/x)$

$\iff e^{-\frac{1}{2}(x-\mu_1)^T \Sigma_1^{-1}(x-\mu_1)} + \ln 0.4 - \frac{1}{2}\ln|\Sigma_1|$

$\geq e^{-\frac{1}{2}(x-\mu_{-1})^T \Sigma_{-1}^{-1}(x-\mu_{-1})} + \ln 0.6 - \frac{1}{2}\ln|\Sigma_{-1}|$

$\iff \underbrace{\frac{1}{2}x^T(\Sigma_1^{-1} - \Sigma_{-1}^{-1})x - x^T(\Sigma_1^{-1}\mu_1 - \Sigma_{-1}^{-1}\mu_{-1}) + \frac{1}{2}\mu_1^T\Sigma_1^{-1}\mu_1 - \frac{1}{2}\mu_{-1}^T\Sigma_{-1}^{-1}\mu_{-1} + \ln\frac{0.6}{0.4} + \frac{1}{2}\ln|\Sigma_1|/|\Sigma_{-1}|}_{q(x)}$

≤ 0

$f^*(x) = \text{sign}(-q(x))$.

Model error $= 0 \iff \exists\, \omega, b \ni \text{sign}(-q(x)) = \text{sign}(\omega^T x + b)$

$\iff$ (i) $q(x) \geq 0 \ \forall x$ &

(ii) $q(x) \leq 0 \ \forall x$ &

(iii) $x^T(\Sigma_1^{-1} - \Sigma_{-1}^{-1})x = 0 \ \forall x$

I

This is because,

if $x^T(\Sigma_1^{-1} - \Sigma_{-1}^{-1})x \neq 0$, for some x, then

$\downarrow$

$L\Lambda L^T$

at least one non-zero λ_i is present. Let $L^T x = v$

$$x^T L \Lambda L^T x = v^T \Lambda v = \sum_{i:\lambda_i \neq 0} \lambda_i v_i^2$$

$$\therefore q(x) \longrightarrow \bar{q}(v) = \sum_{i:\lambda_i \neq 0} \lambda_i v_i^2 + \sum_i \sigma_i v_i + c$$

for some σ, c

$$\omega^T x + b = \bar{\omega}^T v + b$$

$$\mathrm{sign}(\bar{q}(v)) = \mathrm{sign}(\bar{\omega}^T v + b) \quad \forall v.$$

Consider $v = \begin{bmatrix} i \\ \vdots \\ 0 \end{bmatrix} \to i : \lambda_i \neq 0$ some i

$$\mathrm{sign}(\lambda_i v_i^2 + \sigma_i v_i + c) = \mathrm{sign}(\bar{\omega}_i v_i + b)$$

Two sign changes $\to$ Discriminant > 0

zero sign change $\to$ Discriminant ≤ 0

$\bar{\omega}_i = 0 \to$ zero sign change

$\bar{\omega}_i \neq 0 \to$ One sign change.

$\therefore$ (I) is necessary & sufficient.

<u>Analysis of (i)</u> : $q(x) \geq 0 \ \forall x \in \mathbb{R}^n$.

Let $q(x) = \frac{1}{2} x^T A x + \sigma^T x + c$, A is symmetric.

From $\bar{q}$ it is clear $\bar{q}(v) \geq 0 \implies \lambda_i \geq 0 \ \forall i$

else $\bar{q} \to -\infty$ for some v.

$\therefore A \succeq 0$.

$$q(x) = \frac{1}{2}x^TAx + b^Tx + c \geq 0 \quad \forall x$$

$$\iff G(\bar{x},t) = \frac{1}{2}\bar{x}^TA\bar{x} + b^T\bar{x}t + ct^2 \geq 0, \quad \forall \bar{x}, t \neq 0$$

$$(\bar{x}/t = x)$$

$$\iff \frac{1}{2}\bar{x}^TA\bar{x} + b^T\bar{x}t + ct^2 \geq 0 \quad \forall \bar{x}, t$$

(Continuity)

$$\iff \frac{1}{2}\left[\bar{x}^T \; t\right]\begin{bmatrix} A & b \\ b^T & 2c \end{bmatrix}\begin{bmatrix} \bar{x} \\ t \end{bmatrix} \geq 0 \quad \forall \bar{x}, t$$

$$\iff \begin{bmatrix} A & b \\ b^T & 2c \end{bmatrix} \succeq 0$$

$$\iff -\begin{bmatrix} \Sigma_1^{-1} - \Sigma_{-1}^{-1} & \Sigma_{-1}^{-1}\mu_1 - \Sigma_1^{-1}\mu_1 \\ (\Sigma_{-1}^{-1}\mu_{-1} - \Sigma_1^{-1}\mu_1)^T & \mu_1^T\Sigma_1^{-1}\mu_1 - \mu_{-1}^T\Sigma_{-1}^{-1}\mu_{-1} + \ln\frac{0.84|\Sigma_1|}{0.16|\Sigma_{-1}|} \end{bmatrix} \succeq 0$$

$\downarrow$

without -sign $\succeq 0$

Illy for $q(x) \leq 0 \; \forall x$,

Analysis of (iii):

$$x^T(\overbrace{\Sigma_1^{-1} - \Sigma_{-1}^{-1}}^{A})x = 0 \quad \forall x$$

$$\Rightarrow A_{ii} = 0 \quad \forall i \quad \left(\because x = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \to i\text{ entry}\right)$$

Now choose, $x = \begin{array}{c} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \end{array} \begin{array}{l} \\ \\ \rightarrow i \\ \\ \rightarrow j \\ \\ \\ \end{array}$

then $x^T A x = 0 \Rightarrow A_{ii} + A_{jj} + 2A_{ij} = 0$ ($\because A$ is symmetric)

$\Rightarrow A_{ij} = 0$

$\therefore A = 0$. i.e. $\Sigma_1^{-1} = \Sigma_{-1}^{-1} \Leftrightarrow \Sigma_1 = \Sigma_{-1}$.

For linear models sign$(w^T x)$, Case (i) & (ii) remain valid.

Case (iii) now additionally needs c in $q(x)$ to be zero.

i.e. Case (iii): $\Sigma_1 = \Sigma_{-1}$, $\mu_1^T \Sigma_1^{-1} \mu_1 - \mu_{-1}^T \Sigma_{-1}^{-1} \mu_{-1} + \ln \dfrac{0.36|\Sigma_1|}{0.16|\Sigma_{-1}|} = 0$

(3) $\varphi(x_1) = \varphi\left(\begin{bmatrix}0\\0\end{bmatrix}\right) = ((0+0)\%2) - \frac{1}{2} = -\frac{1}{2}$, $y_1 = 1$

$\varphi(x_2) = \varphi\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = ((0+1)\%2) - \frac{1}{2} = \frac{1}{2}$, $y_2 = -1$

$\varphi(x_3) = \varphi\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = ((1+0)\%2) - \frac{1}{2} = \frac{1}{2}$, $y_3 = -1$

$\varphi(x_4) = \varphi\left(\begin{bmatrix}1\\1\end{bmatrix}\right) = ((1+1)\%2) - \frac{1}{2} = -\frac{1}{2}$, $y_4 = 1$

w.l.o.g. the classifier as $\varphi(x)$

is $\text{sign}(\varphi(x)+w_0)$ or $\text{sign}(-\varphi(x)+w_0)$

We can see that $\boxed{\text{sign}(-\varphi(x_i)) = y_i}$ $\forall\, i = 1 \text{ to } 4$

$\boxed{f(x) = \text{sign}(-\varphi(x))}$ $\therefore$ This must be ERM linear classifier as empirical risk with this is zero.

Week 4

① l is smooth $\Leftrightarrow |l'(a,b) - l'(a',b)| \leq L \| a - a' \|$

Now, the objective of risk min. is $h(\omega) \equiv E[l(\omega^T\varphi(x), y)]$.

$$\Rightarrow \nabla h(\omega) = E[l'(\omega^T\varphi(x), y)\varphi(x)]$$

(by chain rule).

$$\Rightarrow \|\nabla h(\omega) - \nabla h(\nu)\| = \| E[l'(\omega^T\varphi(x), y)\varphi(x)] - E[l'(\nu^T\varphi(x), y)\varphi(x)] \|$$

$$= \| E[(l'(\omega^T\varphi(x), y) - l'(\nu^T\varphi(x), y))\varphi(x)] \|$$

$$\leq E[\|(l'(\omega^T\varphi(x), y) - l'(\nu^T\varphi(x), y))\varphi(x)\|]$$

($\because$ Jensen's inequality
$\|\cdot\|$ is convex.)

$$= E[|l'(\omega^T\varphi(x), y) - l'(\nu^T\varphi(x), y)| \, \|\varphi(x)\|]$$

$$\leq E[L|\omega^T\varphi(x) - \nu^T\varphi(x)| \, \|\varphi(x)\|]$$

$$= L E[|(\omega - \nu)^T\varphi(x)| \, \|\varphi(x)\|]$$

$$\leq L E[\|\omega - \nu\| \, \|\varphi(x)\| \, \|\varphi(x)\|]$$

($\because$ Cauchy Schwartz inequality)

$$= \left(L E[\|\varphi(x)\|^2]\right) \|\omega - \nu\|$$

$\therefore$ h is smooth with constant as $L E[\|\varphi(x)\|^2]$.

(2) Consider $g(\omega; x, y) = l(\omega^T \varphi(x), y)$

l is convex $\Leftrightarrow$ $l(\lambda a + (1-\lambda) b, c)$
$\leq \lambda l(a,c) + (1-\lambda) l(b,c)$
$\forall \lambda \in [0,1]$.

$$g(\lambda\omega + (1-\lambda)\nu ; x, y) = l((\lambda\omega + (1-\lambda)\nu)^T \varphi(x), y)$$

$$= l(\lambda\omega^T\varphi(x) + (1-\lambda)\nu^T\varphi(x), y)$$

$$\leq \lambda l(\omega^T\varphi(x), y) + (1-\lambda) l(\nu^T\varphi(x), y)$$

($\because$ l is convex)

$$\cancel{= \lambda g(\omega) + (1-\lambda) g(\nu)}$$

$$= \lambda g(\omega; x, y) + (1-\lambda) g(\nu; x, y)$$

$\therefore$ g is convex,

Consider $h(\omega) = E[g(\omega; x, y)]$

$$= \int g(\omega, x, y)\, dp(x, y)$$

$$h(\lambda\omega + (1-\lambda)\nu) = \int g(\lambda\omega + (1-\lambda)\nu; x, y)\, dp(x, y)$$

$$\leq \int (\lambda g(\omega; x, y) + (1-\lambda) g(\nu; x, y))\, dp(x, y)$$

($\because$ g is convex)

$$= \lambda \int g(\omega; x, y) dp(x, y) + (1-\lambda) \int g(\nu; x, y)\, dp(x, y)$$

$$= \lambda h(\omega) + (1-\lambda) h(\nu) \Rightarrow h \text{ is convex.}$$

Gamma model $\frac{1}{\Gamma(\alpha)} \lambda^{\alpha} y^{\alpha-1} e^{-\lambda y}$

WEEK 6

Poisson model $p_\lambda(y) = \dfrac{\lambda^y e^{-\lambda}}{y!}$, $y = 0, 1, \dots$, $\lambda > 0$

MLE $\rightarrow$ $\underset{\lambda>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log p_\lambda(y_i)$

$= \underset{\lambda>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log\left(\lambda^{y_i} e^{-\lambda} / y_i!\right)$

$= \underset{\lambda>0}{\operatorname{argmax}} \underbrace{\sum_{i=1}^{m} y_i \log \lambda - m\lambda}_{g(\lambda)}$ $\longrightarrow$ concave open domain

$g'(\lambda) = \dfrac{\sum_{i=1}^{m} y_i}{\lambda} - m = 0 \iff \boxed{\hat{\lambda} = \dfrac{\sum_i y_i}{m}}$

Exponential distribution $p_\lambda(y) = \lambda e^{-\lambda y}$, $y > 0$, $\lambda > 0$

MLE $\longrightarrow$ $\underset{\lambda>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log p_\lambda(y_i)$

$= \underset{\lambda>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log\left(\lambda e^{-\lambda y_i}\right)$

$= \underset{\lambda>0}{\operatorname{argmax}} \underbrace{m \log \lambda - \lambda \sum_{i=1}^{m} y_i}_{g(\lambda)}$ $\longrightarrow$ Concave maximization over open domain

$g'(\lambda) = \dfrac{m}{\lambda} - \sum_i y_i = 0 \iff \boxed{\hat{\lambda} = \dfrac{m}{\sum_i y_i}}$

Gamma model $\quad p_{\lambda,\alpha}(z) = \dfrac{\lambda^{\alpha} z^{\alpha-1} e^{-\lambda z}}{\Gamma(\alpha)}$, $z>0$, $\alpha, \lambda > 0$

$$\text{MLE} \rightarrow \underset{\lambda>0,\alpha>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log p_{\lambda,\alpha}(z_i)$$

$$= \underset{\lambda>0,\alpha>0}{\operatorname{argmax}} \sum_{i=1}^{m} \log\left(\frac{\lambda^{\alpha} z_i^{\alpha-1} e^{-\lambda z_i}}{\Gamma(\alpha)}\right)$$

$$= \underset{\lambda>0,\alpha>0}{\operatorname{argmax}} \underbrace{m\alpha \log\lambda + (\alpha-1)\sum_i \log z_i - \lambda \sum_i z_i - m \log\Gamma(\alpha)}_{g(\alpha,\lambda)}$$

$\rightarrow$ convex over domain min

$$\frac{\partial g(\alpha,\lambda)}{\partial\alpha} = m\log\lambda + \sum_i \log z_i - m\frac{\Gamma'(\alpha)}{\Gamma(\alpha)}$$

$\rightarrow$ known as digamma φ

$$= m\log\lambda + \sum_i \log z_i - m\,\varphi(\alpha) = 0$$

$$\frac{\partial g(\alpha,\lambda)}{\partial\lambda} = \frac{m\alpha}{\lambda} - \sum_i z_i = 0$$

Need to solve numerically.

(2) 4.5 (a) $\quad p(x,y\mid\theta_1,\theta_2) = p(y/x,\theta_1,\theta_2)\,p(x/\theta_1,\theta_2)$

$$= p(y/x,\theta_2)\,p(x/\theta_1)$$

) assumption given

	$y=0$	$y=1$
$x=0$	$\theta_2(1-\theta_1)$	$(1-\theta_2)(1-\theta_1)$
$x=1$	$(1-\theta_2)\,\theta_1$	$\theta_2\,\theta_1$

4 (b) MLE $\rightarrow$ $\underset{\substack{\theta_1 \in (0,1) \\ \theta_2 \in (0,1)}}{\operatorname{argmax}} \log(\theta_2 \theta_1) + \log((1-\theta_2)\theta_1) + \log(\theta_2(1-\theta_1))$

$$+ \log((1-\theta_2)\theta_1) + \log(\theta_2 \theta_1)$$

$$+ \log(\theta_2(1-\theta_1)) + \log((1-\theta_2)(1-\theta_1))$$

$$= \underset{\theta_1 \in (0,1)}{\operatorname{argmax}} \; 4\log\theta_2 + 3\log(1-\theta_2)$$

$$+ \underset{\theta_2 \in (0,1)}{\max} \; 4\log\theta_1 + 3\log(1-\theta_1)$$

$$\rightarrow \hat{\theta}_1 = \frac{4}{7}, \quad \hat{\theta}_2 = \frac{4}{7}.$$

$$p(\mathcal{D} \mid \hat{\theta}_1, \hat{\theta}_2) = \left(\frac{4}{7}\right)^2 \left(\frac{3}{7}\frac{4}{7}\right)\left(\frac{3}{7}\frac{4}{7}\right)\left(\frac{3}{7}\frac{4}{7}\right)\left(\frac{4}{7}\right)^2$$

$$\left(\frac{3}{7}\frac{4}{7}\right)\left(\frac{3}{7}\frac{3}{7}\right)$$

$$= \frac{4^8 3^6}{7^{14}}$$

(c) This is 4-class multinoulli so

$$\hat{\theta}_{00} = \frac{2}{7}, \quad \hat{\theta}_{01} = \frac{1}{7}, \quad \hat{\theta}_{10} = \frac{2}{7}, \quad \hat{\theta}_{11} = \frac{2}{7}$$

$$p(\mathcal{D} / \theta) = \left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{2}{7}\right)\left(\frac{1}{7}\right)$$

$$= 2^6 / 7^7$$

(4.7) Murphy

$$E\left[\frac{1}{m}\sum_{i=1}^{m}\left(x_i-\frac{\sum_j x_j}{m}\right)^2\right]$$

$$= E\left[\frac{1}{m}\sum_{i=1}^{m}\left(x_i^2 - 2\frac{\sum_j x_j x_i}{m} + \frac{\sum_{jk} x_j x_k}{m^2}\right)\right]$$

$$= \frac{1}{m}\sum_{i=1}^{m}\left(E\{x_i^2\} - 2\frac{\sum_j E\{x_j x_i\}}{m} + \frac{\sum_{jk} E\{x_j x_k\}}{m^2}\right)$$

$$= E\left[x^2\right] - \frac{1}{m^2}\sum_{ij} E\{x_i x_j\}$$

$$= E\{x^2\} - \frac{1}{m^2}\left(m E\{x^2\} + (m^2-m)\left(E\{x\}\right)^2\right)$$

$$= \left(\frac{m-1}{m}\right)\left(E\{x^2\} - \left(E\{x\}\right)^2\right)$$

$$= \frac{m-1}{m}\sigma^2 \neq \sigma^2 \quad \therefore \text{It is biased.}$$

(3) Multivariate Gaussian:

$$p(x) \propto e^{-\frac{1}{2}(x-\mu)^T\Sigma^{-1}(x-\mu)}$$

$$\propto e^{-\frac{1}{2}x^T\Sigma^{-1}x + \mu^T\Sigma^{-1}x}$$

$$\propto e^{\langle \Sigma^{-1}, -\frac{xx^T}{2}\rangle_F + (\Sigma^{-1}\mu)^T x}$$

$$= e^{\langle w, \varphi(x)\rangle}$$

$$= e$$

$w = (\Sigma^{-1}, \Sigma^{-1}\mu)$

$\varphi(x) = \left(-\frac{xx^T}{2}, x\right)$

$h(x) = 1.$

~~[illegible]~~

Categorical model $p_\theta(z) = \prod_{i=1}^{c} \theta_i^{I\{z=i\}}$

$$= e^{\sum_i I\{z=i\} \log \theta_i}$$

$$w = \begin{bmatrix} \log\theta_1 \\ \vdots \\ \log\theta_c \end{bmatrix}, \quad \varphi(z) = \begin{bmatrix} I\{z=1\} \\ \vdots \\ I\{z=c\} \end{bmatrix} \qquad = e^{w^T\varphi(z)}$$

$\Downarrow$

$$= e^{w^T\varphi(z)}$$

$e^{w_1} + \dots + e^{w_c} = 1$

$$= \frac{e^{w^T\varphi(z)}}{\sum_j e^{w_j}}$$

$h(z) = 1$

Poisson model: $p_\lambda(z) = \frac{\cancel{z}\,\lambda^z e^{-\lambda}}{z!}$, $z \geq 0$, $\lambda > 0$.

$$\propto \frac{1}{z!} e^{z \log \lambda}$$

$\varphi(z) = z$

$w = \log\lambda$

$$= h(z) e^{w^T\varphi(z)}$$

$h(z) = \frac{1}{z!}$

Exponential distribution: $p_\lambda(x) = \lambda e^{-\lambda x}$, $x > 0$, $\lambda > 0$

$$\propto e^{w^T\varphi(x)}$$

$\varphi(x) = x$

$w = -\lambda$

$h(x) = 1$

Gamma model: $p_{\alpha,\lambda}(x) = \frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$, $x > 0$, $\alpha > 0, \lambda > 0$

$$\propto e^{\alpha \log x - \lambda x} \qquad w = \begin{bmatrix} \alpha \\ \lambda \end{bmatrix} > 0$$

$\Rightarrow \varphi(x) = \begin{bmatrix} \log x \\ -x \end{bmatrix}$

$h(x) = 1/x$

(5) $\varphi(y) = \begin{bmatrix} y^2 \\ y^4 \end{bmatrix}$

$$p_{\omega,\gamma}(y) = \frac{e^{\omega y^2 + \gamma y^4}}{Z(\omega,\gamma)}$$

$$Z(\omega,\gamma) = \int_{-\infty}^{\infty} e^{\omega y^2 + \gamma y^4}\, dy$$

$$= 2\int_0^{\infty} e^{\omega y^2 + \gamma y^4}\, dy$$

$\longrightarrow \infty$ if $\cancel{\quad}$ $\gamma > 0$.

$$= \begin{cases} \dfrac{2}{e^{\omega^2/4\gamma}} \displaystyle\int_0^{\infty} e^{\gamma\left(y^2 + \frac{\omega}{2\gamma}\right)^2} dy & , \gamma < 0 \\ 2\displaystyle\int_0^{\infty} e^{\omega y^2} dy = \dfrac{1}{\sqrt{2\pi(-1/2\omega)}} & \text{if } \gamma = 0,\ \omega < 0 \end{cases}$$

Further?

Week 7

7.1

① Generative lin. regression: $\hat{f}(x) = \hat{\Sigma}_{21} \hat{\Sigma}_{11}^{-1} (x - \hat{\mu}_1) + \hat{\mu}_2$.

For lin. regression:

$$\underset{w \in \mathbb{R}^n, w_0 \in \mathbb{R}}{\arg\min} \frac{1}{m} \sum_{i=1}^{m} (w^T x_i + w_0 - y_i)^2$$

$$= \underset{w, w_0}{\arg\min} \underbrace{\left(\frac{1}{m} \sum_{i=1}^{m} w^T x_i x_i^T w + w_0^2 - \frac{2}{m} w^T \left(\sum_i x_i y_i\right) - \frac{2w_0}{m} \sum_i y_i + 2 \frac{w_0}{m} w^T \left(\sum_i x_i\right) \right)}_{g(w, w_0)}$$

$$\nabla_w g(w) = 2 \left(\sum_i \frac{x_i x_i^T}{m} \right) \hat{w} - 2 \frac{\sum_i x_i y_i}{m} + 2\hat{w}_0 \left(\frac{\sum x_i}{m} \right) = 0$$

$$\frac{\partial g(w)}{\partial w_0} = 2\hat{w}_0 - 2 \left(\frac{\sum_i y_i}{m} \right) + 2 \hat{w}^T \left(\frac{\sum_i x_i}{m} \right) = 0$$

Eliminating $\hat{w}_0$,

$$\left(\sum_i \frac{x_i x_i^T}{m} \right) \hat{w} - \frac{\sum_i x_i y_i}{m} + \left(\frac{\sum_i x_i}{m} \right) \left(\frac{\sum_i y_i}{m} - \hat{w}^T \left(\frac{\sum_i x_i}{m} \right) \right) = 0$$

$$\Leftrightarrow \cancel{\left(\sum_i \frac{x_i x_i^T}{m} - \left(\frac{\sum x_i}{m} \right) \right)} \hat{w} = \frac{\sum x_i y_i}{m} - \left(\frac{\sum x_i}{m} \right) \left(\frac{\sum y_i}{m} \right)$$

$$\left(\sum_i \frac{x_i x_i^T}{m} - \left(\frac{\sum_i x_i}{m} \right) \left(\frac{\sum_i x_i}{m} \right)^T \right) \hat{w} =$$

$$\Rightarrow \hat{w} = \hat{\Sigma}_{11}^{-1}\left(\hat{\Sigma}_{12}\right)$$

$$\Rightarrow \hat{w}_0 = \hat{\mu}_2 - \left(\hat{\Sigma}_{21}\hat{\Sigma}_{11}^{-1}\right)\hat{\mu}_1$$

$$\Rightarrow \hat{f}(x) = \hat{w}x + \hat{w}_0$$
$$= \hat{\Sigma}_{21}\hat{\Sigma}_{11}^{-1}(x - \hat{\mu}_1) + \hat{\mu}_2$$

$\therefore$ Even in the case $\hat{\mu}_1$, $\hat{\mu}_2$ are not zero, the final predictive functions are exactly the same.

~~② Let $\hat{\Sigma}_{2i}$ and $\hat{\mu}_{2i}$ denote the sample mean and~~

~~Let $X =$~~

② From the final form $\hat{f}(x) = \hat{\Sigma}_{21}\hat{\Sigma}_{11}^{-1}(x - \hat{\mu}_1) + \hat{\mu}_2$ it is clear that the i^{th} component of $\hat{f}(x)$ is only a function of i^{th} row of $\hat{\Sigma}_{21}$ and i^{th} entry of $\hat{\mu}_2$.

$$\left(\hat{f}(x)\right)_i = \underbrace{\left(\hat{\Sigma}_{21}\right)_i}_{i^{th}\text{ row}}\hat{\Sigma}_{11}^{-1}(x - \hat{\mu}_1) + \left(\hat{\mu}_2\right)_i$$

To estimate $(\hat{\Sigma}_{21})_i$ and $(\hat{\mu}_2)_i$ we need only i^{th} output from the training set

Unless $\hat{\Sigma}_{22}$ is involved, this decoupling issue will remain!

③ Here, $p(x,y) = p(x/y)\, p(y)$

$p(x/y) \sim N(W^T y + b, \Sigma_1)$, $p(y) \sim N(\mu_2, \Sigma_2)$

$p(y/x) \propto p(x/y)\,p(y)$ ($\because$ Bayes rule)

$$\propto e^{-\frac{1}{2}(x - W^T y - b)^T \Sigma_1^{-1} (x - W^T y - b)}\, e^{-\frac{1}{2}(y-\mu_2)^T \Sigma_2^{-1}(y-\mu_2)}$$

$$\propto e^{-\frac{1}{2} y^T W \Sigma_1^{-1} W^T y - \frac{1}{2} y^T \Sigma_2^{-1} y + y^T W \Sigma_1^{-1}(x-b) + y^T \Sigma_2^{-1}\mu_2}$$

$$= e^{-\frac{1}{2} y^T (W\Sigma_1^{-1}W^T + \Sigma_2^{-1}) y + y^T (W\Sigma_1^{-1}(x-b) + \Sigma_2^{-1}\mu_2)}$$

$$\propto e^{-\frac{1}{2}\left[y - (W\Sigma_1^{-1}W^T + \Sigma_2^{-1})^{-1}(W\Sigma_1^{-1}(x-b) + \Sigma_2^{-1}\mu_2)\right]^T (W\Sigma_1^{-1}W^T + \Sigma_2^{-1}) \left[y - (W\Sigma_1^{-1}W^T + \Sigma_2^{-1})^{-1}(W_1\Sigma_1^{-1}(x-b) + \Sigma_2^{-1}\mu_2)\right]}$$

$$\therefore f(x) = E[y/x] = (W\Sigma_1^{-1}W^T + \Sigma_2^{-1})^{-1}(W_1\Sigma_1^{-1}(x-b) + \Sigma_2^{-1}\mu_2)$$

$$KL(p^*(x,y)\,\|\,p_\theta(x,y)) = E\left[KL(p^*(x/y)\,\|\,p_\theta(x/y))\right] + KL(p^*(y)\,\|\,p_\theta(y))$$

decoupling of objective

$\hat{\mu}_2, \hat{\Sigma}_2$

Sample mean & covariance of y values.

$$\operatorname*{argmin}_{W,b,\Sigma_1} E\left[KL\left(\hat{p}(x/y) \,\|\, p_{W,b,\Sigma_1}(x/y)\right)\right]$$

$$= \operatorname*{argmax}_{W,b,\Sigma_1} E\left[\log p_{W,b,\Sigma_1}(x/y)\right]$$

$$= \operatorname*{argmax}_{W,b,\Sigma_1} \frac{1}{m}\sum_{i=1}^{m} -\frac{1}{2}(x_i - W^T y_i - b)^T \Sigma_1^{-1}(x_i - W^T y_i - b) - \frac{1}{2}\log|\Sigma_1|$$

For fixed Σ_1, problem wrt W, b:

$$\operatorname*{argmin}_{W,b} \sum_i (W^T y_i - (x_i - b))^T \Sigma_1^{-1} (W^T y_i - (x_i - b))$$

$$\bar{W} = W \Sigma_1^{-1/2}$$
$$\bar{b} = \Sigma_1^{-1/2} b$$
$$\bar{x}_i = \Sigma_1^{-1/2} x_i$$

$$\operatorname*{argmin}_{\bar{W},\bar{b}} \sum_{i=1} (\bar{W}^T y_i + \bar{b} - \bar{x}_i)^T (\bar{W}^T y_i + \bar{b} - \bar{x}_i)$$

$$\bar{W} = \hat{\Sigma}_{22}^{-1} \bar{\Sigma}_{21}, \quad \bar{b} = \bar{\mu}_1 - \bar{\Sigma}_{12} \hat{\Sigma}_{22}^{-1} \hat{\mu}_2$$

$$\hat{W} = \hat{\Sigma}_{22}^{-1} \hat{\Sigma}_{21}, \quad \hat{b} = \hat{\mu}_1 - \hat{\Sigma}_{12} \hat{\Sigma}_{22}^{-1} \hat{\mu}_2$$

(independent of Σ_1).

For fixed, $\hat{W}$, b, Σ_1 is ~~covariance~~ correlation sample with data $\hat{W}^T y_i - x_i + \hat{b}$

4

~~$f(x) = \Sigma_{22}^{-1}\Sigma_{21}\Sigma_{1}^{-1}\Sigma$~~ (Same derivation as Week 6 (1))

$$\rightarrow \hat{\Sigma}_1 = \frac{1}{m}\sum_i (\hat{W}^T y_i - x_i + \hat{b})(\hat{W}^T y_i - x_i + \hat{b})^T$$

Final regressor: $\hat{f}(x) = (\hat{W}\hat{\Sigma}_1^{-1}\hat{W}^T + \hat{\Sigma}_2^{-1})^{-1}(\hat{W}\hat{\Sigma}_1^{-1}(x - \hat{b}) + \hat{\Sigma}_2^{-1}\hat{\mu}_2)$

Observe constant term: $(\hat{W}\hat{\Sigma}_1^{-1}\hat{W}^T + \hat{\Sigma}_2^{-1})^{-1}\hat{\Sigma}_2^{-1}\hat{\mu}_2$

$$= (\hat{\Sigma}_{22}^{-1}\hat{\Sigma}_{21}\hat{\Sigma}_1^{-1}\hat{\Sigma}_{12}\Sigma_{22}^{-1} + \hat{\Sigma}_2^{-1})^{-1}\hat{\Sigma}_2^{-1}\hat{\mu}_2$$

which does not decompose into individual regressors for each output. Hence decoupling issue is finally resolved!

~~(4) $p(x|y) \sim N(\mu_y, \sigma_y^2)$~~

7.2 (1) ~~$p(y|x)$~~ Poisson likelihood with mean λ is

$$p(k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, \ldots$$

So, $$p_w(y|x) = \frac{(e^{w^T\varphi(x)})^y e^{-e^{w^T\varphi(x)}}}{y!}, \quad y = 0, 1 \ldots$$

MLE: $\underset{w}{\operatorname{argmax}} \frac{1}{m} \sum_{i=1}^{m} \log P_w(y_i/x_i)$

$$= \underset{w}{\operatorname{argmax}} \frac{1}{m} \sum_{i=1}^{m} \log\left(\frac{\left(e^{w^T\varphi(x_i)}\right)^{y_i} e^{-e^{w^T\varphi(x_i)}}}{y_i\,!} \right)$$

$$= \underset{w}{\operatorname{argmax}} \frac{1}{m} \underbrace{\sum_{i=1}^{m} \left[y_i \left(w^T\varphi(x_i)\right) - e^{w^T\varphi(x_i)} \right]}_{h(w)}$$

$$\nabla h(w) = \sum_{i=1}^{m} \left[\cancel{\frac{y_i}{w^T\varphi(x_i)}}\; y_i\,\varphi(x_i) - e^{w^T\varphi(x_i)}\,\varphi(x_i) \right]$$

The solution can be found by running SGD:

$$w^{(k+1)} = w^{(k)} - s_k \left[y_k\,\varphi(x_k) - e^{w^{(k)T}\varphi(x_k)}\,\varphi(x_k) \right]$$

Let solution be $\hat{w}$.

0-1 loss: $\hat{f}(x) = \lfloor e^{\hat{w}^T\varphi(x)} \rfloor$ (mode of Poisson with parameter $e^{\hat{w}^T\varphi(x)}$)

abs. dev. loss: $\hat{f}(x) \rightarrow$ number between $e^{\hat{w}^T\varphi(x)} - \ln 2$ and $e^{\hat{w}^T\varphi(x)} + \frac{1}{3}$.

(median of Poisson)

(2) Though, this is a generative model, there may be no samples of x for a particular y so as $y \in \mathbb{N}$.

$\therefore$ parameters μ, σ^2 are to be found using: (like in discriminative MLE)

$$\underset{\mu,\sigma^2}{\operatorname{argmax}} \frac{1}{m} \sum_{i=1}^{m} \log p(x_i/y_i)$$

$$= \underset{\mu,\sigma^2}{\operatorname{argmax}} \frac{1}{m} \sum_{i=1}^{m} \log\left(\frac{e^{-\frac{1}{2}\frac{(x_i - \mu y_i)^2}{2\sigma^2 y_i^2}}}{\sqrt{2\pi\sigma^2 y_i^2}} \right)$$

$$= \underset{\mu,\sigma^2}{\operatorname{argmax}} -\frac{1}{2\sigma^2} \sum_{i=1}^{m} \left(\frac{x_i - \mu y_i}{y_i^2}\right)^2 - \frac{1}{2} \sum_{i=1}^{m} \log \sigma^2 y_i^2$$

Let $\lambda = 1/\sigma^2 > 0$

$$\Rightarrow = \underset{\lambda>0,\, \mu}{\operatorname{argmin}} \frac{\lambda}{2} \sum_{i=1}^{m} \left(\frac{x_i - \mu y_i}{y_i^2}\right)^2 - \frac{1}{2} \sum_{i=1}^{m} \log\left(\lambda / y_i^2\right)$$

Problem wrt. μ is least squares with data as x_i/y_i.

$$\therefore \boxed{\hat{\mu} = \frac{1}{m} \sum_i \frac{x_i}{y_i}}$$

Problem wrt λ is $\underset{\lambda>0}{\operatorname{argmin}}\, g(\lambda)$, $g(\lambda) = \frac{\lambda}{2} \sum_i \left(\frac{x_i - \hat{\mu} y_i}{y_i}\right)^2 - \frac{m \log \lambda}{2}$

$\downarrow$

convex over open set.

∴ We wish to find $\hat{\delta} \ni g'(\hat{\delta}) = 0$

$$\Leftrightarrow \frac{1}{2}\sum_i \left(\left(\frac{x_i}{y_i}\right) - \hat{u}\right)^2 - \frac{m}{2\hat{\delta}} \cancel{\sum_i \log(\ldots)} = 0$$

$$\Leftrightarrow \boxed{\hat{\delta} = \frac{\cancel{\sum_i \log(\ldots)}\, m}{\sum_i \left(\frac{x_i}{y_i} - \hat{u}\right)^2}} > 0$$

~~$p(y)$ is standard Poisson~~:

$$\Leftrightarrow \sigma^2 = \frac{\sum_i \left(\hat{u} - x_i/y_i\right)^2}{m}$$

Problem wrt λ is MLE for Poisson:

$$\hat{\lambda} = \frac{\sum_{i=1}^{m} (y_i - 1)}{m}$$

→ Week ⑥ ① from.

$$p(y|x) \propto p(x/y)\, p(y)$$

$$= \frac{e^{-\frac{(x - \hat{u} y)^2}{2\hat{\sigma}^2 y}}}{\sqrt{y}} \cancel{(y-1)} \frac{\hat{\lambda}^{(y-1)} e^{-\hat{\lambda}}}{(y-1)!}$$

→ Not Poisson!

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$$\hat{f}(x) = \underset{y \in \mathbb{N}}{\operatorname{argmax}} -\frac{(x-\hat{\mu} y)^2}{2\hat{\sigma}^2 y} + (y-1)\log\hat{\lambda} - \log\left(\sqrt{y}\,(y-1)!\right)$$

I do not know how to solve this ;)

WEEK 9

① $k(x,y) \equiv \cos(x-y)$

$= \cos x \cos y + \sin x \sin y$

$= \varphi(x)^T \varphi(y)$, where $\varphi(x) = \begin{bmatrix} \cos x \\ \sin x \end{bmatrix}$

② $k(x,y) \equiv \cos(h(x) - h(y))$

$= \cos(h(x)) \cos(h(y)) + \sin(h(x)) \sin(h(y))$

$= \varphi(x)^T \varphi(y)$, where $\varphi(x) = \begin{bmatrix} \cos(h(x)) \\ \sin(h(x)) \end{bmatrix}$

③ $\cosh(k(x,y)) = \sum_{i=0}^{\infty} \frac{k(x,y)^{2i}}{(2i)!}$

Consider $k_n(x,y) = \frac{k(x,y)^{2n}}{(2n)!}$. k_n is kernel as product of kernels is a kernel by theorem 29.0.2

Consider $\bar{k}_n(x,y) = k_0(x,y) + \dots + k_n(x,y)$

$\bar{k}_n$ is kernel as sum of kernels is kernel by

$\cosh(k(x,y)) = \lim_{n\to\infty} \bar{k}_n(x,y) \rightarrow$ kernel by

(4) h is kernel $\Rightarrow h(x,y) = \langle \varphi(x), \varphi(y) \rangle$

$$\Rightarrow k(X,Y) = \iint f_X(x) f_Y(y)\, h(x,y)\, dx\, dy$$

$$= \iint f_X(x) f_Y(y) \langle \varphi(x), \varphi(y) \rangle\, dx\, dy$$

$$= \iint \langle \varphi(x) f_X(x), \varphi(y) f_Y(y) \rangle\, dx\, dy$$

$$= \langle \underbrace{\int \varphi(x) f_X(x)\, dx}_{\bar{\varphi}(X)}, \int \varphi(y) f_Y(y)\, dy \rangle$$

$$= \langle \bar{\varphi}(X), \bar{\varphi}(Y) \rangle$$

$\therefore$ k is a kernel.